

Comprehensive Health Playbook

A biopsychosocial assessment framework + evidence map for outcomes that matter

Version 2026-03-04 • Prepared for technical readers (medicine + statistics + engineering)

Scope: determinants, measurement, physiology, pathology, prevention, and interventions.

How to use this document (and what it is not)

This playbook is a modular map of the variables that shape human health and the ways we can measure and influence them. It is designed for building comprehensive assessments, longitudinal monitoring programs, and evidence-based intervention plans.

Not medical advice. It is educational and research-oriented. Decisions should be individualized, guideline-aligned, and coordinated with qualified clinicians.

Design principles

- Think in **systems**: context → behaviors & care → physiology → outcomes (with feedback loops).
- Prefer **trajectories** over one-off values; measure change and variability.
- Use **tiered testing**: start low-burden/high-signal; add higher-cost tests only if they change decisions.
- Treat evidence as a stack: mechanistic plausibility ≠ clinical benefit; prioritize RCTs/meta-analyses for outcomes.
- Avoid low-value cascades: false positives, overdiagnosis, and opportunity costs are real harms.

Working definition of health

WHO's constitutional definition frames health as “complete physical, mental and social well-being,” not merely absence of disease. [1]

Table of Contents

How to use this document (and what it is not)

2

Design principles

2

Working definition of health

2

Table of Contents

3

1. A systems view of health

6

Determinants vs mechanisms vs outcomes

6

Function and disability are part of health

6

2. Measurement architecture (what to collect)

7

What to optimize

7

3. Evidence map: how to compare interventions

8

Minimal evidence rubric

8

Selected levers with strong outcome associations

8

Global burden context

8

4. Social determinants, environment, and resources

9

Interventions

9

5. Mental health, cognition, and psychosocial functioning

10

Evidence-based levers (selected)

10

6. Sleep as a health behavior (and a diagnostic domain)

11

What to measure

11

Interventions (high-level)

11

7. Diet and nutrition

12

High-signal patterns and outcomes

12

Practical measurement

12

8. Physical activity, fitness, and function

13

Targets (guideline-aligned, simplified)

13

Measurement

13

9. Tobacco, nicotine, and other substances

14

Assessment

14

Interventions

14

10. Relationships, purpose, and community

15

Measurement

15

Interventions

15

11. Core clinical measures: vitals, anthropometrics, function

16

12. Laboratory biomarkers (core + add-ons)

17

Outcome-linked pathways	17
13. Cardiovascular system	18
14. Metabolic and endocrine system	19
Key measurements	19
Key levers	19
15. Respiratory system and sleep-disordered breathing	20
16. Kidney health	21
17. Liver and gastrointestinal health	22
18. Hematology, immune function, and infections	23
19. Neurologic and sensory health	24
20. Musculoskeletal health, pain, bone, and mobility	25
21. Dermatologic health and sun exposure	26
22. Sexual and reproductive health	27
23. Preventive care: screening and immunizations	28
24. Cancer risk, early detection, and low-value testing	29
25. Pharmacology: medication optimization and supplement skepticism	30
Medication review checklist	30
26. Wearables, digital biomarkers, and omics (optional layer)	31
27. Integrating everything: a technical workflow for personal health systems	32
Workflow	32
10a. Work, stress, and recovery	33
What to measure	33
Interventions	33
10b. Oral health	34
Assessment	34
Interventions	34
10c. Injury prevention and safety	35
Measurement	35
Interventions	35
20a. Aging, frailty, and longevity science (brief)	36
Frailty / function-first approach	36
Appendix D. Data dictionary template (starter)	37
Appendix E. Extended test menu by domain (choose only with a question)	38
Appendix F. Causal inference cheat sheet (for personal health experiments)	39
Appendix A. Comprehensive intake prompts (biopsychosocial)	40
A1. Goals, constraints, and baseline	40

A2. Environment and SDOH 40

A3. Mental health and stress 40

Appendix B. Measurement checklist (minimal dashboard -> extended) 41

B1. Minimal weekly dashboard (example) 41

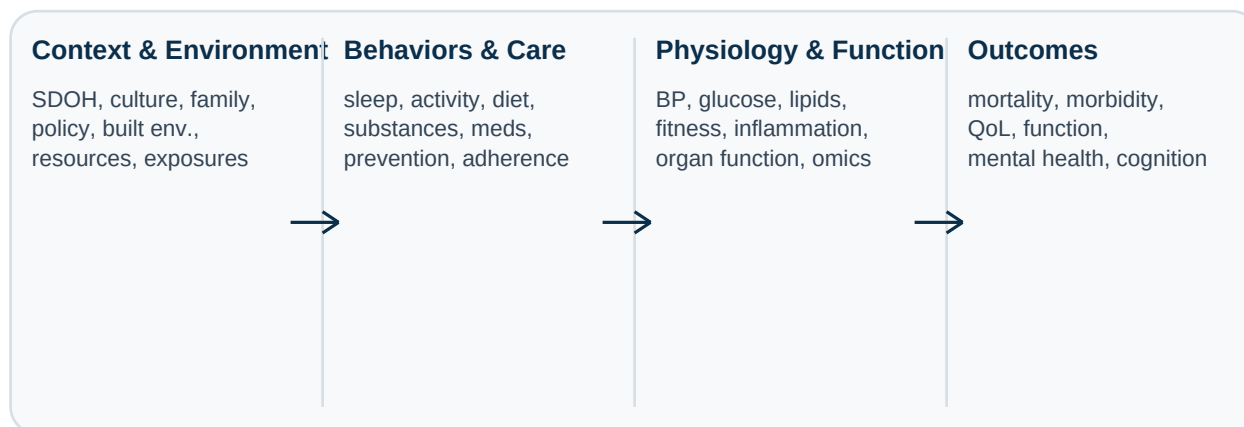
B2. Quarterly / semiannual dashboard (example) 41

B3. Extended diagnostics (only with a question) 41

Appendix C. References (selected, high-load-bearing) 42

1. A systems view of health

A practical health model is a directed graph: context and environment shape behaviors and access to care; behaviors and care alter physiology and function; physiology drives outcomes (morbidity, mortality, quality of life).



Determinants vs mechanisms vs outcomes

- **Determinants:** social determinants, culture, built environment, education, resources, exposures, trauma, policies. [2,3]
- **Mechanisms:** inflammation, insulin sensitivity, vascular function, autonomic tone, neuroplasticity, immune function.
- **Outcomes:** survival, incident disease, disability, symptoms, functioning, cognition, mood, and life satisfaction.

Function and disability are part of health

The WHO ICF framework explicitly includes functioning and environmental factors, complementing disease classifications. [4]

2. Measurement architecture (what to collect)

Measure what you can act on. Use a tiered architecture and keep a clean data dictionary.

Tier	Typical tools	Examples	Primary use
0. Context	Structured interview + records	Family history, housing, work, finances, exposures	Risk stratification; feasibility
1. Patient-reported outcomes	PROMIS and validated scales [5]	Depression, anxiety, pain, fatigue, sleep	Symptoms and QoL tracking
2. Vitals + anthropometrics	BP, HR, waist, body composition	Home BP, waist-to-height, DEXA (optional)	Cardiometabolic risk; trends
3. Basic labs	CBC, CMP, lipids, HbA1c	Glucose, LDL-C, renal function	Detect disease; guide prevention
4. Function tests	Fitness + performance batteries	VO2/CRF, gait speed, grip strength	Healthspan and prognosis
5. Targeted diagnostics	Imaging + specialty tests	CAC, echo, spirometry, sleep study	Answer clinical questions
6. Continuous monitoring	Wearables + CGM (selected)	Steps, sleep, HRV, glucose patterns	Feedback loops; anomaly detection
7. Omics (optional)	Genetics, proteomics, metabolomics	Polygenic risk, pharmacogenomics	Niche decisions; hypothesis generation

What to optimize

- **Healthspan:** physical and cognitive function over time (ICF framing). [4]
- **Well-being:** patient-reported physical, mental, and social health (PROMIS framing). [5]
- **Risk:** future disease and death (vitals, labs, fitness).

3. Evidence map: how to compare interventions

For each lever, capture population, dose, comparator, outcomes, effect size, and certainty. Prefer outcomes that matter (mortality, events, disability, QoL) over surrogate-only evidence.

Minimal evidence rubric

- **Effect size:** absolute + relative (risk difference, HR/RR, NNT when possible).
- **Outcome proximity:** clinical outcomes > validated PROs > surrogates.
- **Bias risk:** confounding, reverse causality, selection bias, publication bias.
- **Harm profile:** adverse events, overdiagnosis, false positives, opportunity cost.

Selected levers with strong outcome associations

Lever / exposure	Outcome(s)	Effect size (illustrative)	Ref
Physical activity (~150 min/wk moderate)	All-cause & CVD mortality	RR ~0.69 all-cause at 8.75 mMET-h/wk	[6]
Higher cardiorespiratory fitness (CRF)	All-cause mortality	High vs low: HR ~0.47; per +1 MET: ~11–17% lower mortality	[7]
Resistance training	All-cause mortality	Max benefit around ~60 min/wk: RR ~0.74	[8]
Exercise for major depression	Depression symptoms	Walking/jogging g ~-0.62; yoga -0.55; strength -0.49 vs active controls	[9]
Smoking cessation	Mortality; life expectancy	Current vs never: HR ~2.7–2.8; cessation <3y can avert ~5 life-years lost vs continued smoking	[32]
BP lowering (per -5 mmHg SBP)	Major CV events	~10% lower risk per 5 mmHg SBP reduction	[12]
LDL lowering (per -1 mmol/L)	Major vascular events	~22% lower events; ~10% lower all-cause mortality	[13]
Mediterranean diet (PREDIMED)	Major CV events	HR ~0.69 (EVOO) and 0.72 (nuts) vs control	[14]
>=7000 steps/day	Incident depression	RR ~0.69 vs <7000; +1000 steps: RR ~0.91	[10]
Alcohol	Population burden	GBD: minimum risk at 0; WHO: no safe threshold for carcinogenic effects	[15,16]

Global burden context

Global Burden of Disease analyses highlight the outsized contribution of risks such as particulate air pollution, high systolic BP, smoking, and high fasting plasma glucose. [30,31]

4. Social determinants, environment, and resources

SDOH shape health opportunities and constraints: safety, food access, stress load, and ability to adhere to care. They are core to the WHO framing and to Healthy People 2030 priorities. [2,3]

Domain	Examples of variables	Common downstream pathways
Economic stability	Income volatility, debt stress, benefits, job security	Chronic stress, access, adherence
Education & literacy	Health literacy, language access	Safety, self-management
Built environment	Walkability, safety, transport, heat, pollution	Activity opportunity, exposures, injuries
Food environment	Food security, cooking resources, UPF reliance	Diet quality, metabolic risk
Social context	Isolation, discrimination, caregiving load	Mood, physiologic stress, mortality association
Healthcare access	Insurance, continuity, trust, time barriers	Prevention, chronic disease control

Interventions

- Resource navigation: transport, food support, housing stabilization, benefits enrollment.
- Work design: schedule control, ergonomic changes, burnout reduction, reasonable workloads.
- Social prescribing: structured community connection, group exercise, volunteering.
- Environmental mitigation: indoor air quality, heat mitigation, safety planning.

5. Mental health, cognition, and psychosocial functioning

Mental health is a primary outcome and a mediator of every other health behavior. Assess symptoms, functioning, risk, and context.

Construct	Examples (not exhaustive)	Notes
Depression	PHQ-9; PROMIS Depression [5]	USPSTF recommends screening with adequate supports. [24]
Anxiety	GAD-7; PROMIS Anxiety [5]	Often co-occurs with insomnia and substance use
Sleep disturbance	PROMIS Sleep [5]	Sleep is a major health behavior. [26,27,29]
Trauma/PTSD	PCL-5	Consider safety and stabilization first
Substance use	AUDIT-C; DAST-10	Alcohol and nicotine have major population impact. [15,16,32]
Cognition	MoCA; NIH Toolbox	Track change over time; consider hearing and sleep

Evidence-based levers (selected)

Exercise has RCT evidence as an effective depression treatment, with walking/jogging, yoga, and strength training showing moderate symptom reductions in a large network meta-analysis. [9]

- Psychotherapy: CBT, interpersonal therapy, exposure-based therapies (indication-dependent).
- Medication: antidepressants/anxiolytics when indicated; monitor side effects and interactions.
- Sleep: treat insomnia (CBT-I) and screen for sleep apnea when indicated; aim for ≥ 7 hours for most adults. [26,27]

6. Sleep as a health behavior (and a diagnostic domain)

Sleep is a driver of mood, cardiometabolic health, cognition, and safety. CDC and AASM/SRS emphasize that most adults should obtain about 7 or more hours per night. [26,27]

What to measure

- Duration (weekday/weekend), consistency, latency, awakenings, subjective quality.
- Daytime sleepiness, napping, performance errors, driving risk.
- Snoring/witnessed apneas, morning headaches (sleep apnea screen).

Interventions (high-level)

- Sleep opportunity: consistent schedule, light exposure timing, noise/temperature control.
- Behavioral: CBT-I, stimulus control, sleep restriction (clinician-guided).
- Medical: evaluate for sleep apnea, restless legs, circadian disorders when suspected.

7. Diet and nutrition

Diet is best evaluated as patterns, not single nutrients. Prioritize adequacy (protein, fiber, micronutrients), then optimize cardiometabolic risk factors using evidence-based patterns.

High-signal patterns and outcomes

Pattern / lever	Primary outcomes	Evidence snapshot	Ref
Mediterranean pattern (PREDIMED style)	Major CV events	Lower events vs control in high-risk adults (HR ~0.69–0.72)	[14]
DASH pattern	Blood pressure	Meta-analysis: mean SBP reduction about -3 mmHg vs control	[11]
Reducing ultra-processed foods	All-cause mortality (observational)	Higher UPF intake associated with higher mortality; interpret with confounding	[34]
Alcohol minimization	Cancer and overall burden	GBD risk curves: minimum risk at 0; WHO: no safe threshold for carcinogenic effects	[15,16]

Practical measurement

- 24-hour recall snapshots (repeat) or food frequency questionnaires for pattern-level scoring.
- Track a few sentinel variables: fiber intake, protein distribution, sugary beverage frequency, UPF share.
- Biomarker proxies (selective): HbA1c, triglycerides, ApoB/LDL, BP respond to diet changes.

8. Physical activity, fitness, and function

Activity and fitness have broad associations with mortality, cardiovascular disease, cancer, mood, and function. The biggest gains often occur moving from inactive to modest activity. [6,7]

Targets (guideline-aligned, simplified)

The Physical Activity Guidelines for Americans (2nd ed.) provide evidence-based targets for aerobic and muscle-strengthening activity. [28]

- Aerobic: build toward ~150+ min/week moderate (or equivalent) as a practical baseline.
- Strength: 2+ sessions/week with progressive overload; dose-response evidence suggests benefits even at low volumes. [8]
- Reduce sedentary time: frequent breaks and 'movement snacks'.

Measurement

- Steps/day and minutes of moderate-to-vigorous activity; consider wearables for trends.
- CRF (VO₂ estimate): treadmill/bike test, step test, or validated non-exercise equations. [7]
- Function tests: sit-to-stand, gait speed, balance, grip strength.

9. Tobacco, nicotine, and other substances

Combustible tobacco is among the highest-risk exposures. In pooled cohort analyses, current smoking is associated with ~2.7–2.8x hazard for death vs never smoking; quitting yields large benefits, visible within a few years. [32]

Assessment

- Current and historical use, dependence, triggers, prior quit attempts, readiness to quit.
- Alcohol pattern (frequency, binge episodes), cannabis and other drugs, impacts on function and mood.
- Co-use with sleep or anxiety symptoms (common reinforcing loops).

Interventions

- Nicotine cessation: behavioral support + pharmacotherapy when indicated (NRT, varenicline, bupropion).
- Alcohol: brief intervention, motivational interviewing; treat comorbid anxiety/depression; consider medication when indicated.
- Harm reduction and safety planning: overdose prevention where relevant; avoid mixing sedatives with alcohol.

10. Relationships, purpose, and community

Social connection is a measurable determinant with a long literature linking it to health outcomes; a large meta-analysis found social relationships predict mortality risk in magnitude comparable to established risk factors. [17]

Measurement

- Network size and frequency of contact, perceived support, loneliness, conflict, and relationship safety.
- Role meaning/purpose, volunteering, community participation.

Interventions

- Structured connection: clubs, group exercise, volunteering, peer support.
- Relationship skills and therapy when needed; address safety first in abusive contexts.
- Treat hearing loss and sleep problems that can amplify isolation.

11. Core clinical measures: vitals, anthropometrics, function

These measures are cheap, repeatable, and often more predictive than exotic biomarkers when tracked longitudinally.

Measure	What it captures	Strengths	Caveats / notes
Blood pressure	Hemodynamic load	Direct pathway to CV events; home BP useful for confirmation [33]	Technique matters; consider ambulatory BP
Waist / waist:height	Central adiposity	Often better signal than BMI	Track trends; measurement consistency
DEXA (selected)	Fat/lean + bone	Sarcopenia/osteoporosis /athletes	Not routine; decision-dependent
CRF estimate	Integrated oxygen transport	Strong predictor across outcomes [7]	Requires protocol consistency
Strength/function	Healthspan proxy	Grip, sit-to-stand, gait speed	Interpret with age and comorbidity

12. Laboratory biomarkers (core + add-ons)

Labs are valuable when they change decisions. Use a core panel to detect common pathology, then add targeted tests for specific hypotheses.

Category	Typical tests	Primary use
Hematology	CBC with differential	Anemia, infection signals, hematologic abnormalities
Metabolic	CMP (electrolytes, renal, liver), fasting glucose or HbA1c	Renal/hepatic function, metabolic disease
Lipids	Total, HDL, triglycerides, LDL-C	ASCVD risk estimation; therapy monitoring
Urine (selected)	Urinalysis; albumin/creatinine ratio when indicated	Kidney disease, diabetes complications

Outcome-linked pathways

- Blood pressure lowering reduces major cardiovascular events (~10% per -5 mmHg SBP). [12]
- LDL lowering reduces major vascular events (~22% per -1 mmol/L). [13]
- Diet/activity/sleep shape these pathways upstream. [6,11,14,26,27]

13. Cardiovascular system

AHA Life's Essential 8 summarizes core cardiovascular health metrics: diet, physical activity, nicotine, sleep, BMI, blood lipids, blood glucose, and blood pressure. [29]

Test	What it answers	When commonly used	Caveats
Resting ECG	Rhythm/conduction	Symptoms, known disease	Low yield as blanket screening
Home/ambulatory BP	True BP load	Confirm diagnosis; therapy titration	Technique and device validation matter
Echo	Structure/function	Murmur, HF symptoms, known disease	Not routine screening
Stress testing	Ischemia/functional capacity	Chest pain eval; risk stratification	False positives in low-risk populations
CAC	Plaque burden	Risk reclassification when uncertain	Radiation; should change management

14. Metabolic and endocrine system

Metabolic dysfunction is a central pathway to cardiometabolic disease, kidney disease, neuropathy, and cognitive decline.

Key measurements

- Glycemia: fasting glucose and/or HbA1c; diabetes screening is USPSTF B in adults 35–70 with overweight/obesity. [22]
- Body composition and central adiposity: waist measures and trends.
- Lipids and BP: core cardiometabolic risk pathways with strong outcome evidence. [12,13]

Key levers

- Physical activity and fitness improvements. [6,7]
- Dietary patterns (Mediterranean/DASH) and UPF reduction. [11,14,34]
- Sleep adequacy and regularity. [26,27]

15. Respiratory system and sleep-disordered breathing

Pulmonary health interacts with activity capacity, sleep quality, cardiovascular risk, and overall quality of life.

- Core: symptoms + smoking exposure history; spirometry when asthma/COPD suspected or known.
- Sleep apnea suspicion: snoring, witnessed apneas, daytime sleepiness; consider sleep study when indicated.
- High-leverage interventions: smoking cessation and improved fitness/activity. [6,7,32]

16. Kidney health

Kidney health is both an outcome and a mediator: it affects BP regulation, electrolyte balance, medication clearance, and cardiovascular risk.

- Measure: eGFR (creatinine), urine albumin/creatinine ratio when indicated, BP control. [33]
- Manage upstream risks: hypertension and diabetes are major drivers of CKD progression. [12,22]
- Medication review: dose adjustment and nephrotoxin avoidance.

17. Liver and gastrointestinal health

GI and hepatic health affect nutrition, inflammation, energy, and medication tolerance.

- Assess: reflux, bowel habit changes, bleeding, weight loss, abdominal pain, alcohol use pattern.
- Labs (selected): liver enzymes, bilirubin; consider viral hepatitis risk assessment.
- Colorectal cancer screening is one of the highest-value preventive interventions. [20]

18. Hematology, immune function, and infections

Immune health includes baseline disease susceptibility, vaccine-mediated protection, inflammatory state, and iatrogenic risks.

- Core labs: CBC for anemia/infection signals; interpret in clinical context.
- Vaccinations: follow the current CDC adult immunization schedule for age and risk group. [25]
- Lifestyle supports immune function indirectly (sleep, stress, nutrition, activity). [26,27,28]

19. Neurologic and sensory health

Brain and nerve health includes cognition, mood, pain, motor/sensory function, and autonomic regulation.

- Track cognition longitudinally; focus on vascular risk control (BP, lipids, diabetes) and sleep/activity. [12,13,26,28]
- Peripheral neuropathy workup when indicated: diabetes, B12, thyroid, alcohol, meds.
- Vision and hearing: early correction supports function and social connection.

20. Musculoskeletal health, pain, bone, and mobility

Musculoskeletal capacity is a major driver of healthspan: independence, injury risk, and activity capacity.

- Assess: pain, injuries, weakness, mobility, falls, and work/ergonomic contributors.
- Intervene: progressive resistance training; combine with balance and aerobic work. [8,28]
- Bone health: consider DEXA when risk factors for osteoporosis; guideline- and risk-based.

21. Dermatologic health and sun exposure

Skin health affects cancer risk, autoimmune disease burden, infections, and quality of life.

- Assess: changing lesions, chronic wounds, severe eczema/psoriasis, occupational exposures.
- Sun protection behaviors: UV avoidance, sunscreen, protective clothing.
- Targeted dermatology evaluation for suspicious lesions rather than routine broad tumor-marker testing.

22. Sexual and reproductive health

Sexual health includes function, safety, consent, fertility goals, contraception, and STI prevention.

- Assess: menstrual history, sexual function concerns, pain, libido, erectile function, fertility intentions.
- STI risk assessment and prevention counseling; vaccination where appropriate (e.g., HPV).
- Mental health, sleep, and vascular/metabolic health are common mediators of sexual function.

23. Preventive care: screening and immunizations

Preventive care aims to reduce risk factors and detect disease early when treatment changes outcomes.

Condition	Who (simplified)	Test / approach	Interval (typical)	Ref
Hypertension	Adults ≥ 18	Office BP + confirm outside clinic	Yearly if ≥ 40 /high risk	[33]
Depression	All adults (with supports)	Validated screening tool	Periodic	[24]
Prediabetes / T2D	Adults 35–70 with OW/OB	Fasting glucose or HbA1c	Depends on risk/results	[22]
Breast cancer	Women 40–74	Mammography	Biennial	[19]
Colorectal cancer	Adults 45–75	Several modalities	Varies	[20]
Lung cancer	50–80 + 20 pack-yr; quit < 15 y	Annual low-dose CT	Annual while eligible	[21]

For additional topics and nuanced details, see the USPSTF A/B recommendations index. [18]

Immunizations change; consult the current CDC adult schedule (updated regularly). [25]

24. Cancer risk, early detection, and low-value testing

Evidence-based screening focuses on tests with proven net benefit. Avoid broad 'tumor marker panels' and non-indicated imaging in asymptomatic low-risk people: false positives and overdiagnosis can cause harm.

- Use proven screenings (breast, colorectal, lung in eligible groups). [19,20,21]
- Use risk-based genetics when actionable, with counseling and clear follow-up plans.
- Lifestyle pathways matter: nicotine, alcohol, adiposity, activity, and diet quality. [15,16,32]

25. Pharmacology: medication optimization and supplement skepticism

Medication review is a mandatory component of comprehensive health. Supplements should be treated like drugs: benefit-harm and evidence quality matter.

Medication review checklist

- Indication: current, evidence-based reason for each medication.
- Dose and organ function: renal/hepatic adjustment; age-related pharmacokinetics.
- Interactions: drug-drug and drug-supplement; prioritize high-risk combinations.
- Adherence and usability: complexity, cost, side effects, preferences.
- Deprescribing opportunities: coordinate with clinicians.

26. Wearables, digital biomarkers, and omics (optional layer)

Wearables can improve adherence and provide trend data, but they measure proxies. Omics can reveal risk and biology, but actionability is often limited.

- Wearables: steps, sleep estimates, resting HR, training load; validate with reality (e.g., home BP, symptoms).
- CGM: useful for selected metabolic questions; avoid over-interpretation of noise.
- Genetics: focus on high-penetrance actionable variants and pharmacogenomics; polygenic risk is probabilistic.

27. Integrating everything: a technical workflow for personal health systems

Transform high-dimensional data into a small set of priorities, tracked over time, with explicit hypotheses and feedback loops.

Workflow

- **Define outcomes:** longevity, healthspan (function), mood, cognition, performance.
- **Establish baseline:** context + behaviors + vitals/labs + function + PROs. [2,3,5]
- **Choose bottlenecks:** nicotine, BP, activity/fitness, sleep, depression often dominate EV. [6,7,12,32]
- **Instrument lightly:** a small dashboard beats a huge panel.
- **Run experiments:** change one lever at a time; measure adherence and effect size.
- **Review cadence:** weekly (behaviors), quarterly (function/labs as needed), annually (prevention and major review).

10a. Work, stress, and recovery

Work and chronic stress shape sleep opportunity, diet quality, activity capacity, substance use risk, and mental health. A practical assessment focuses on demands, control/autonomy, schedule stability, recovery time, and meaning.

What to measure

- Hours, shift work, commute time, predictability, after-hours work, on-call load.
- Perceived control/autonomy, role clarity, psychological safety, burnout symptoms.
- Recovery: sleep opportunity, off-days, vacations, social time, physical activity feasibility.

Interventions

- Schedule engineering: protect sleep windows; reduce rotating shifts when possible.
- Work redesign: increase autonomy, reduce unnecessary load, improve ergonomics, reduce exposure hazards.
- Stress skills: mindfulness/CBT-based tools, therapy when needed; address the upstream stressors first.

10b. Oral health

Oral health affects pain, nutrition, sleep, and (via periodontal disease) may relate to cardiometabolic risk pathways. It is often missed in 'medical-only' health models.

Assessment

- Dental pain, bleeding gums, dry mouth (medications), grinding (stress/sleep).
- Preventive care access: cleanings, fluoride exposure, diet pattern (sugary beverages).

Interventions

- Daily hygiene (brushing, flossing/interdental cleaning), fluoride, regular dental care.
- Address sleep apnea/bruxism contributors and reflux when present.

10c. Injury prevention and safety

Injuries are a major source of disability and can derail long-term health trajectories. Prevention is largely about environment, behavior, and strength/balance capacity.

Measurement

- Falls history, balance confidence, vision/hearing, sedating meds, alcohol use.
- Occupational and recreational hazards (vehicles, tools, sports).

Interventions

- Strength + balance training; footwear; home safety checks; vision correction.
- Medication review to reduce sedation/falls; alcohol risk reduction.
- Protective equipment and safety culture in high-risk activities.

20a. Aging, frailty, and longevity science (brief)

Longevity (lifespan) and healthspan (functional years) diverge. Aging biology is an active research field; clinically, the highest-leverage, most evidence-backed levers remain: blood pressure control, atherogenic lipid lowering when indicated, smoking cessation, physical activity/fitness, sleep, mental health, and social connection.

Frailty / function-first approach

- Track function: gait speed, sit-to-stand, balance, grip strength, unintentional weight loss, fatigue.
- Prioritize protein adequacy, resistance training, fall prevention, and medication simplification.
- Avoid over-testing; focus on interventions that preserve independence and cognition.

Appendix D. Data dictionary template (starter)

A data dictionary keeps your longitudinal dataset clean. Below is a starter template; expand per your needs.

Variable	Unit / scale	Frequency	Source	Notes
BP_systolic_home	mmHg	2–7x/week (if relevant)	Validated cuff	Average of 2–3 readings; same arm/time
Steps_per_day	count	daily	Wearable/phone	Use 7-day rolling average
Sleep_duration	hours	daily	Wearable + self-report	Track schedule consistency
Mood_score	0–10 or PROMIS	weekly	Self-report	Define anchor points
HbA1c	%	3–12 months (if relevant)	Lab	Use same lab method when possible
LDL_C	mg/dL or mmol/L	6–12 months	Lab	Consider ApoB for discordance
CRF_estimate	METs	quarterly/biannual	Fitness test	Use consistent protocol
Waist_circumference	cm	weekly/monthly	Tape measure	Standardize landmark

Appendix E. Extended test menu by domain (choose only with a question)

This list is intentionally broad. The key discipline is to select tests only when the result changes a decision.

Domain	Examples of tests	Typical decision use
Cardiovascular	ECG, echo, stress testing, CAC, ambulatory BP	Diagnose symptoms; refine prevention strategy
Metabolic	OGTT, fasting insulin (selected), CGM, liver ultrasound (selected)	Clarify dysglycemia; guide lifestyle/meds
Pulmonary	Spirometry, diffusion capacity, sleep study	Asthma/COPD confirmation; sleep apnea treatment
Bone/MSK	DEXA, vitamin D (selected), imaging for red flags	Fracture risk; treatment selection
Cancer (risk-based)	Genetic panels, targeted imaging	High-risk syndromes; symptom evaluation
Neuro	Neuropsych testing, MRI (symptoms)	Evaluate cognitive decline; exclude secondary causes

Appendix F. Causal inference cheat sheet (for personal health experiments)

- Prefer randomized trials for efficacy; use observational studies for real-world associations and rare harms.
- Beware regression to the mean: re-test after extreme values before overreacting.
- Control for seasonality (sleep/activity) and life events; use rolling averages for noisy measures.
- When running self-experiments: define baseline, intervention, outcomes, adherence metrics, and stopping rules.
- Use minimally sufficient adjustment sets; avoid adjusting for mediators when estimating total effects.

Appendix A. Comprehensive intake prompts (biopsychosocial)

Use this as a menu, not a script. Start with what is most relevant, and expand only as needed.

A1. Goals, constraints, and baseline

- Goals (healthspan, longevity, performance, symptom relief) and time horizon.
- Constraints: finances, time, social context, access to care, risk tolerance.
- Past medical history, meds, allergies, surgeries, family history, prior labs/imaging.

A2. Environment and SDOH

- Housing stability, neighborhood safety/noise/heat, indoor air quality, transport.
- Food security and cooking resources; UPF reliance; alcohol exposure in social context.
- Work demands, shift work, autonomy, ergonomics, occupational exposures.
- Social support, loneliness, discrimination, caregiving load.

A3. Mental health and stress

- Mood/anxiety/trauma symptoms, cognition, attention, substance use patterns.
- Sleep schedule and insomnia, sleep apnea signals, daytime sleepiness.
- Safety: suicidality and violence exposure where relevant.

Appendix B. Measurement checklist (minimal dashboard -> extended)

B1. Minimal weekly dashboard (example)

- Sleep duration + quality; schedule consistency. [26,27]
- Activity: steps/day; 2x/week strength exposure if feasible. [6,8]
- Mood: brief weekly rating or PROMIS short form. [5]
- BP: home readings when elevated risk or diagnosed HTN. [33]

B2. Quarterly / semiannual dashboard (example)

- Weight + waist trend; function test (sit-to-stand, gait speed).
- Fitness estimate (submax test), grip strength.
- Labs as indicated: HbA1c/glucose, lipid panel/ApoB, renal function.

B3. Extended diagnostics (only with a question)

- Sleep study, spirometry, CAC scoring, targeted imaging, specialty labs.
- Define the decision rule in advance: 'If X, we will do Y.'

Appendix C. References (selected, high-load-bearing)

Curated list favoring primary guidelines, large meta-analyses, and open-access sources.

- [1] World Health Organization. Constitution of the World Health Organization (definition of health). <https://www.who.int/about/governance/constitution>
- [2] World Health Organization. Social determinants of health (fact sheet). <https://www.who.int/news-room/fact-sheets/detail/social-determinants-of-health>
- [3] U.S. Office of Disease Prevention and Health Promotion (ODPHP). Healthy People 2030: Social Determinants of Health. <https://odphp.health.gov/healthypeople/priority-areas/social-determinants-health>
- [4] World Health Organization. International Classification of Functioning, Disability and Health (ICF). <https://www.who.int/standards/classifications/international-classification-of-functioning-disability-and-health>
- [5] NIH Common Fund. PROMIS (Patient-Reported Outcomes Measurement Information System). <https://commonfund.nih.gov/promis>
- [6] Garcia L, Pearce M, Abbas A, et al. Non-occupational physical activity and risk of cardiovascular disease, cancer and mortality outcomes: dose-response meta-analysis. *Br J Sports Med.* 2023;57(15):979-989. PMID: 36854652. <https://pubmed.ncbi.nlm.nih.gov/36854652/>
- [7] Lang JJ, Prince SA, et al. Cardiorespiratory fitness is a strong and consistent predictor of morbidity and mortality: overview of meta-analyses. *Br J Sports Med.* 2024. PMCID: PMC11103301. <https://pmc.ncbi.nlm.nih.gov/articles/PMC11103301/>
- [8] Shailendra P, Baldock KL, Li Y, et al. Resistance training and mortality risk: a systematic review and meta-analysis. *Am J Prev Med.* 2022. PMID: 35599175. <https://pubmed.ncbi.nlm.nih.gov/35599175/>
- [9] Noetel M, Sanders T, et al. Effect of exercise for depression: network meta-analysis of RCTs. *BMJ.* 2024;384:e075847. PMCID: PMC10870815. <https://pubmed.ncbi.nlm.nih.gov/38355154/>
- [10] Bizzozero-Peroni B, et al. Daily Step Count and Depression in Adults: systematic review and meta-analysis. *JAMA Netw Open.* 2024. <https://jamanetwork.com/journals/jamanetworkopen/fullarticle/2828073>
- [11] Filippou CD, et al. DASH diet and blood pressure: systematic review and meta-analysis. *Adv Nutr.* 2020. PMID: 32330233. <https://pubmed.ncbi.nlm.nih.gov/32330233/>
- [12] Rahimi K, et al. Pharmacological blood pressure lowering: individual patient-level meta-analysis. *Lancet.* 2021. PMID: 33933205. <https://pubmed.ncbi.nlm.nih.gov/33933205/>
- [13] Cholesterol Treatment Trialists' (CTT) Collaboration. Intensive LDL lowering and vascular outcomes: meta-analysis of 26 trials. *Lancet.* 2010. PMID: 21067804. <https://pubmed.ncbi.nlm.nih.gov/21067804/>
- [14] Estruch R, Ros E, et al. Mediterranean diet for primary prevention of cardiovascular disease (PREDIMED, corrected and republished). *N Engl J Med.* 2018. PMID: 29897866. <https://pubmed.ncbi.nlm.nih.gov/29897866/>
- [15] Griswold MG, et al. Alcohol use and burden for 195 countries and territories, 1990-2016. *Lancet.* 2018. [https://www.thelancet.com/article/S0140-6736\(18\)31310-2/fulltext](https://www.thelancet.com/article/S0140-6736(18)31310-2/fulltext)
- [16] World Health Organization (Europe). No level of alcohol consumption is safe for our health. 2023. <https://www.who.int/europe/news/item/04-01-2023-no-level-of-alcohol-consumption-is-safe-for-our-health>
- [17] Holt-Lunstad J, Smith TB, Layton JB. Social relationships and mortality risk: meta-analytic review. *PLoS Med.* 2010;7(7):e1000316. <https://journals.plos.org/plosmedicine/article?id=10.1371/journal.pmed.1000316>
- [18] U.S. Preventive Services Task Force (USPSTF). A and B Recommendations. <https://www.uspreventiveservicestaskforce.org/uspstf/recommendation-topics/uspstf-a-and-b-recommendations>
- [19] USPSTF. Breast Cancer: Screening (Final, Apr 30, 2024). <https://www.uspreventiveservicestaskforce.org/uspstf/recommendation/breast-cancer-screening>

- [20] USPSTF. Colorectal Cancer: Screening (May 18, 2021).
<https://www.uspreventiveservicestaskforce.org/uspstf/recommendation/colorectal-cancer-screening>
- [21] USPSTF. Lung Cancer: Screening (Mar 9, 2021).
<https://www.uspreventiveservicestaskforce.org/uspstf/recommendation/lung-cancer-screening>
- [22] USPSTF. Prediabetes and Type 2 Diabetes: Screening (Aug 24, 2021).
<https://www.uspreventiveservicestaskforce.org/uspstf/recommendation/screening-for-prediabetes-and-type-2-diabetes>
- [23] USPSTF. Statin Use for the Primary Prevention of CVD (Aug 23, 2022).
<https://www.uspreventiveservicestaskforce.org/uspstf/recommendation/statin-use-in-adults-preventive-medication>
- [24] USPSTF. Depression and Suicide Risk in Adults: Screening (Jun 20, 2023).
<https://www.uspreventiveservicestaskforce.org/uspstf/recommendation/screening-depression-suicide-risk-adults>
- [25] CDC. Adult Immunization Schedule by Age (updated Oct 7, 2025; includes 2025-2026 COVID-19 guidance).
<https://www.cdc.gov/vaccines/hcp/imz-schedules/adult-age.html>
- [26] CDC. About Sleep (recommended sleep by age; updated May 15, 2024). <https://www.cdc.gov/sleep/about/index.html>
- [27] Watson NF, et al. Recommended amount of sleep for a healthy adult: AASM/SRS consensus statement. J Clin Sleep Med. 2015. <https://aasm.org/resources/pdf/adultsleepdurationconsensus.pdf>
- [28] U.S. HHS. Physical Activity Guidelines for Americans, 2nd edition (2018).
https://odphp.health.gov/sites/default/files/2019-09/Physical_Activity_Guidelines_2nd_edition.pdf
- [29] Lloyd-Jones DM, et al. Life's Essential 8: Updating cardiovascular health metrics. Circulation. 2022.
<https://www.ahajournals.org/doi/10.1161/CIR.0000000000001078>
- [30] GBD 2021 Risk Factors Collaborators. Global burden and strength of evidence for 88 risk factors, 1990-2021. Lancet. 2024;403:2162-2203. PMID: PMC11120204. <https://pubmed.ncbi.nlm.nih.gov/38762324/>
- [31] Australian National University (research portal). Summary of top risk factors contributing to DALYs in 2021 from GBD 2021 risk factor paper (PM2.5, high SBP, smoking, etc.).
<https://researchportalplus.anu.edu.au/en/publications/global-burden-and-strength-of-evidence-for-88-risk-factors-in-204/>
- [32] Cho ER, et al. Smoking Cessation and Short- and Longer-Term Mortality. NEJM Evidence. 2024. PMID: 38329816.
<https://pubmed.ncbi.nlm.nih.gov/38329816/>
- [33] USPSTF. Hypertension in Adults: Screening (Apr 27, 2021).
<https://www.uspreventiveservicestaskforce.org/uspstf/recommendation/hypertension-in-adults-screening>
- [34] Fang Z, et al. Association of ultra-processed food consumption with all-cause mortality. BMJ. 2024;385:e078476.
<https://www.bmj.com/content/385/bmj-2023-078476>

Health Measurement Technologies & Procedures Atlas

Devices, tests, and procedures to quantify human health - from non-contact sensing to invasive diagnostics

Version 3.0 - March 04, 2026

Educational use only. Not medical advice. Costs are approximate U.S. cash-pay ranges and vary by location, provider, and insurance. Regulatory status and product availability change over time; verify current labeling and local availability before purchase or use.

Table of Contents

- 1. Measurement mindset: invasiveness, accuracy, and utility
- 2. Non-contact and wearable measurement
- 3. Home specimen tests (urine, saliva, stool, capillary blood, swabs, breath)
- 4. Clinic-based labs and point-of-care testing
- 5. Imaging technologies (ultrasound, X-ray, CT, MRI, nuclear imaging) + screening imaging
- 6. Functional diagnostics (cardiopulmonary, neuromuscular, pulmonary function, balance, cognition)
- 7. Invasive diagnostics (endoscopy, biopsy, lumbar puncture, catheterization, implants)
- 8. Building a measurement stack (minimalist to advanced)
- 9. Emerging and controversial measurement technologies
- References

1. Measurement mindset: invasiveness, accuracy, and utility

Health measurement is a design problem: every test is a trade between (a) information gained, (b) burden and risk, (c) cost, and (d) actionability. A good measurement program uses the **least invasive** method that is accurate enough to change decisions, and escalates only when the expected value of information outweighs downside.

1.1 A practical invasiveness ladder

Level	Typical inputs	Examples	Key risks / limitations
0 - Non-contact	Questionnaires, smartphone logs, environment	Validated surveys; symptom diaries; digital phenotyping	Bias, missingness, behavior change due to monitoring
1 - External non-invasive	Surface sensors, cuffs, optics	Upper-arm BP cuff; smartwatch HR/PPG; thermometer	Motion artifacts; device validation varies
2 - Non-invasive specimens	Naturally excreted samples	Urine dipsticks; saliva tests; stool FIT	Pre-analytics (timing, contamination) drives error
3 - Minimally invasive	Capillary blood / shallow insertion	Fingerstick glucose/ketones; nasal swabs; CGM filament insertion	Pain; infection (rare); calibration/lag effects
4 - Moderately invasive	Venipuncture, IV contrast, sedation	Blood draw panels; contrast CT/MRI; sedation dentistry	Bleeding; contrast reactions; sedation risks
5 - Invasive procedures	Endoscopy, needle biopsy, puncture	Colonoscopy; liver biopsy; lumbar puncture	Perforation/bleeding; infection; post-procedure complications
6 - Highly invasive	Catheter-based / implant / surgery	Coronary angiography; implantable loop recorder; surgery	Procedure complications; hospital stay; higher cost

1.2 Three validity layers

Analytical validity: does the device measure the signal correctly (accuracy, precision, drift)? **Clinical**

validity: does the measured signal correspond to the clinical state of interest (sensitivity/specificity)?

Clinical utility: does acting on the result improve outcomes (benefit > harm)?

Consumer health products often have good usability but weaker clinical validity or utility. When stakes are high (e.g., oxygen saturation, arrhythmia detection, cancer screening), prefer devices/tests that have been independently validated and have clear follow-up pathways.

2. Non-contact and wearable measurement

Wearables and at-home devices excel at frequent, low-friction measurement. They are most useful for **trends** (baseline shifts, variability, response to interventions) rather than single-point diagnosis - unless the device is specifically cleared/validated for a diagnostic use case.

2.1 Blood pressure (BP)

For home monitoring, the American Heart Association recommends an automatic, cuff-style, **upper-arm** monitor; wrist and finger monitors are generally less reliable. Choose a monitor that has been validated (e.g., via ValidateBP.org). [1-3]

Modality	How it works	Home/clinic	Typical cost	Notes (accuracy, workflow)
Upper-arm oscillometric cuff	Measures oscillations during cuff deflation; estimates SBP/DBP	Home + clinic	Device purchase: ~\$40-\$200 (varies)	Best-supported home method. Use correct cuff size and standardized technique. [1,2]
Ambulatory BP monitor (ABPM)	Automated cuff inflations every 15-30 min for 24 h	Clinic-prescribed (take-home)	Often hundreds of dollars; varies by system/provider	Useful to confirm hypertension and assess nocturnal BP patterns (white-coat/masked HTN). [4]
Cuffless BP (emerging)	Optical PPG + algorithms (pulse wave analysis); may require calibration	Consumer wearables	Example: Hilo Band estimated ~\$280 (launch pricing) [45]	Convenient and frequent, but validation and calibration requirements matter. [45-47]

2.2 Heart rate, rhythm, and cardiac auscultation

Wearables estimate heart rate using photoplethysmography (PPG) - measuring changes in reflected light as blood volume pulses. Electrocardiograms (ECG/EKG) directly measure cardiac electrical potentials via electrodes. Digital stethoscopes measure acoustic energy (heart sounds) using microphones and signal processing; some also add a 1-lead ECG.

Tool	Signal	Typical use	Typical cost	Safety / limitations
Smartwatch PPG (HR, HRV proxies)	Optical pulse waveform	Resting HR trends; activity intensity; recovery signals	Device purchase: ~\$100-\$800+ (brand/model dependent)	Motion artifacts; HRV metrics vary by vendor and sampling.
Apple Watch ECG app	Single-lead ECG (Lead I)	On-demand rhythm recording; AFib checks	Requires Apple Watch Series 4+ (watch cost varies)	ECG app checks for AFib; FDA De Novo clearance exists for the ECG app classification. [5-7]
KardiaMobile (AliveCor)	Single-lead ECG	On-demand ECG with phone; AFib screening	Device purchase: ~\$79 [9]	FDA-cleared personal ECG; requires proper hand placement; not a full diagnostic workup. [8,9]
KardiaMobile 6L (AliveCor)	6-lead ECG	More detailed rhythm characterization than single-lead	Device purchase: ~\$129 [9]	Still not a 12-lead ECG; useful for clinician review. [9]

Tool	Signal	Typical use	Typical cost	Safety / limitations
Digital stethoscope + ECG (Eko DUO-type)	Heart sounds + 1-lead ECG	Remote monitoring and telehealth triage; clinician review	Device purchase varies by model; Eko lists ECG-enabled stethoscopes in the ~\$299-\$449 range (example). [68,69]	Useful adjunct; not a replacement for echocardiography or formal cardiology evaluation when indicated.
Patch ECG (Zio-type) / Holter	Continuous multi-hour/day ECG	Arrhythmia detection with higher yield than spot checks	Procedure: ~\$120-\$378 (Holter, cash-pay examples) [32]	Skin irritation possible; data quality depends on wear time and adherence.

2.3 Oxygenation and breathing

Pulse oximeters estimate oxygen saturation (SpO₂) via red/infrared light absorption. They can be useful, but the FDA notes limitations and a risk of inaccuracy under certain circumstances, including performance differences across skin tones. [10,11]

Tool	Signal	Use case	Typical cost	Important limitations
Finger pulse oximeter (SpO ₂)	Optical absorbance	Spot check oxygenation	Device purchase: ~\$15-\$80+	Accuracy can degrade with poor perfusion, motion, nail polish, and skin pigmentation; interpret clinically. [10,11]
Home spirometer / peak flow meter	Airflow and volume	Asthma/COPD monitoring; pulmonary function trends	Device purchase: ~\$30-\$200+	Technique-sensitive; best for within-person trends.
Capnography (EtCO ₂) (usually clinical)	Exhaled CO ₂	Ventilation monitoring in sedation/critical care	Clinical device/procedure	Usually not home-use; requires proper sampling interface.

2.4 Sleep: from actigraphy to polysomnography

Consumer wearables estimate sleep via actigraphy (movement) plus heart-rate/PPG features; they are useful for sleep timing and regularity. Clinical sleep studies measure respiratory events, oxygenation, sleep stages, and arousals with higher fidelity.

Approach	What it measures	Setting	Typical cost	Notes
Wearable sleep tracking	Movement + HR/PPG-derived features	Home	Included with wearable device	Good for schedules/trends; sleep stage accuracy varies by device.
Home Sleep Apnea Test (HSAT)	Breathing effort/airflow, SpO ₂ ; limited channels	Home	~\$150-\$500 out of pocket (typical); MDSave range ~\$146-\$620 [29,48]	Best for suspected obstructive sleep apnea in otherwise healthy adults; not as comprehensive as lab study. [48]

Approach	What it measures	Setting	Typical cost	Notes
In-lab polysomnography	EEG sleep staging, airflow, effort, ECG, SpO2, limb movement	Sleep lab	Often ~\$1,000-\$3,000+; MDsave example ~\$1,585 (regional) [30,49]	Most comprehensive; enables complex diagnoses and PAP titration. [49]

2.5 Body composition and musculoskeletal metrics

Body weight is easy to measure; body composition is harder. Bioelectrical impedance (BIA) scales estimate fat/lean mass using impedance models. DEXA uses dual-energy X-ray absorptiometry to estimate bone mineral density and can provide body composition estimates in some settings.

Tool	Measures	Setting	Typical cost	Safety / limitations
Smart scale (BIA)	Weight; impedance-derived body fat % (model-based)	Home	\$20-\$200+	Hydration and timing affect estimates; best for trend monitoring.
DEXA (bone density)	Bone mineral density (BMD); sometimes body comp	Clinic / imaging center	MDsave range ~\$64-\$432 [27]	Very low radiation dose: effective dose typically <1-15 microSv for spine+femur exam (IAEA). [17]
Strength: dynamometer	Grip strength; sometimes limb strength	Home/clinic	\$20-\$400+	Grip strength correlates with health outcomes but is technique-sensitive.
Gait / balance (wearables, force plates)	Step variability, sway, ground reaction forces	Home (wearable) / lab	Consumer wearables vs lab systems	Lab systems offer richer biomechanics; consumer devices give coarse proxies.

2.6 Glucose and metabolic monitoring

Glucose can be measured in urine (coarse, threshold-based), capillary blood (fingerstick), or continuously via subcutaneous sensors (CGM) that sample interstitial glucose.

Method	Specimen	Home/clinic	Typical cost	Notes
Urine glucose strip	Urine	Home	\$5-\$20 per pack	Detects glycosuria above renal threshold; not a substitute for blood glucose.
Fingerstick glucometer	Capillary blood	Home	\$10-\$50 meter + strips	Useful for spot checks and calibration; technique and strip quality matter.
CGM (Dexcom G7)	Interstitial glucose (subcutaneous sensor)	Home (prescription)	GoodRx: pricing varies; examples show sensors/receivers in the hundreds without insurance. [20,21]	Trends + alerts; interstitial lag vs blood. Confirm unexpected readings with fingerstick when symptoms disagree.
CGM (FreeStyle Libre 3)	Interstitial glucose (subcutaneous sensor)	Home (prescription)	GoodRx: sensor pricing roughly in the ~\$150 range (varies). [22]	Abbott issued a device correction for some Libre 3/3 Plus sensors due to risk of falsely low readings. [23]

Safety note: CGM systems are generally safe but are still medical devices. Follow manufacturer instructions, watch for skin irritation or infection at insertion sites, and treat any device alarms or inconsistencies seriously.

2.7 Fitness and performance measurement tools

Performance measurement ranges from low-cost proxies (steps, pace) to lab-grade physiology (lactate, gas exchange). Accuracy requirements depend on your goal: trend tracking vs training prescription vs clinical evaluation.

Tool	Measures	Where used	Typical cost	Notes
Chest-strap HR monitor	Electrical heart signal (ECG-derived RR intervals)	Training / home	\$50-\$150	Often more accurate than wrist PPG during vigorous movement.
Power meter (cycling) / running power	Mechanical power output (strain gauges or IMU models)	Training	\$200-\$1,500+	Direct work output; valuable for pacing and load quantification.
Handheld lactate meter	Blood lactate (fingerstick)	Training / sports lab	\$200-\$500 meter + strips	Useful for threshold estimation; requires protocol discipline.
Metabolic analyzer (VO2 testing)	O2/CO2 exchange via mask	Sports lab / clinic	Typically offered as \$100-\$500+ tests (provider dependent). [59,60]	Gold standard for VO2max/thresholds; protocol matters.

2.8 Temperature and autonomic proxies (stress/arousal)

Common consumer sensors include skin temperature, electrodermal activity (EDA, a proxy for sympathetic arousal), and accelerometry-based respiration estimates. These are best interpreted as trend signals rather than diagnostic endpoints.

Tool	Signal	Use case	Typical cost	Notes
Digital oral/tympanic thermometer	Body temperature	Fever detection	\$10-\$60	Spot checks; technique matters (especially tympanic).
Wearable skin temperature	Skin temp trend	Cycle tracking, illness trend	Included with some wearables	Skin temperature is not core temperature; environmental effects matter.
Wearable EDA (galvanic skin response)	Skin conductance	Arousal/stress trend	Included with select wearables	Context-sensitive; movement and contact quality affect signal.

3. Home specimen tests (urine, saliva, stool, capillary blood, swabs, breath)

Home specimen testing spans simple colorimetric strips to lab-grade assays using mailed samples. The main advantages are convenience and frequency. The main failure modes are pre-analytic: timing, contamination, storage/temperature, and user technique.

3.1 Urine testing

Urine testing is low-cost and non-invasive. Most home tests are reagent-strip based: a chemical pad changes color as analytes react (pH, protein, glucose, ketones, nitrites, etc.).

Test / device	What it measures	Use cases	Typical cost	Notes / limitations
Urine dipstick (multi-parameter)	Glucose, ketones, protein, blood, nitrite, leukocyte esterase, pH, specific gravity	Screening for UTI signals, ketones, kidney warning signs	\$5-\$25 per bottle/box	False positives/negatives common; confirm clinically when symptomatic.
Urine albumin (microalbumin) kits	Albumin-to-creatinine ratio proxies	Kidney risk screening (diabetes/HTN)	Varies; often clinic-based	Quantitative ACR is usually best done in a lab for accuracy.
Pregnancy test	hCG in urine (immunoassay)	Pregnancy detection	\$1-\$15+	Highly sensitive near/after missed period; timing matters.
Ovulation (LH) strips	LH surge detection (immunoassay)	Fertility timing	\$10-\$30 per pack	Inter-individual variability; interpret alongside cycle patterns.

3.2 Capillary blood (fingerstick) testing

Fingerstick tests use a lancet to obtain capillary blood. Most modern meters are electrochemical: enzymes on a strip react with the analyte and generate an electrical current proportional to concentration.

Test	What it measures	Typical consumer options	Typical cost	Notes
Glucose	Current blood glucose	Glucometer + strips	Meters ~\$10-\$50; strips often ~\$0.40-\$0.70 each (brand dependent). [51]	Good for spot checks and to confirm CGM discrepancies.
Ketones (beta-hydroxybutyrate)	Blood ketones	Ketone meter + strips	Meters ~\$20-\$60; strips ~\$0.50-\$2 each (varies)	More specific than urine ketone strips for ketoacidosis risk monitoring.
At-home HbA1c kits	Approx. 3-month average glycemia	Mailed kit or rapid kit	Out-of-pocket often starts around ~\$50 per kit. [50]	Turnaround varies from days to weeks; confirm with a lab for clinical decisions.

3.3 Stool testing

Stool tests can screen for occult blood, inflammation, infection, or cancer-associated DNA. They range from inexpensive immunochemical cards to higher-cost lab assays.

Test	What it measures	Typical cost	When used	Notes
FIT (fecal immunochemical test)	Human hemoglobin in stool (antibody-based)	DTC example ~\$69 (mail-in); OTC pricing varies. [41]	Colorectal cancer screening (annual in many strategies)	Positive results require follow-up colonoscopy. Screening intervals depend on strategy. [44]
Stool DNA-FIT (e.g., Cologuard-type)	Hemoglobin + DNA markers	Example: self-pay option listed at \$790 for Cologuard Plus. [54]	Colorectal cancer screening (every 1-3 years depending on guideline)	More expensive; still requires colonoscopy if positive. [44,54]
Stool pathogen PCR panels (lab)	Bacterial/viral/parasitic nucleic acids	Often hundreds of dollars (lab-billed)	Acute infectious diarrhea in select cases	High sensitivity; interpretation depends on clinical context.
Microbiome profiling (consumer)	16S/shotgun sequencing composition	\$100-\$400+ (varies)	Research/wellness	Clinical utility still evolving; avoid over-interpreting single reports.

3.4 Swabs and infectious disease tests

Rapid antigen tests detect viral proteins (typically lateral-flow immunoassays) and trade sensitivity for speed and convenience. Molecular tests (PCR/NAAT) detect nucleic acids and are usually more sensitive but require specialized instrumentation or a lab.

Test type	Examples	Specimen	Typical cost	Notes
OTC rapid antigen	COVID-19 antigen home tests	Nasal swab	Example: iHealth lists ~\$15.69 (pricing varies). [36]	FDA maintains a list of authorized at-home OTC COVID tests. [35]
OTC molecular (where available)	Home COVID/flu molecular systems	Nasal swab	Higher cost than antigen (varies by brand)	Closer to lab NAAT performance but requires dedicated reader/consumables.
OTC HIV self-test	OraQuick oral swab HIV test	Oral fluid swab	Retail example: ~\$29.99-\$39.88 (varies). [37]	Detects HIV antibodies; recent exposures may not be detected until antibodies develop. Confirm positives clinically.
OTC syphilis antibody test	First To Know Syphilis Test	Fingerstick blood	Marketed at 'under \$30' and reported around \$29.98. [38,39]	Treponemal antibody test: positive may indicate current or past infection; requires confirmatory testing and clinician follow-up. [38,39]

Test type	Examples	Specimen	Typical cost	Notes
Home sample collection NAAT	Chlamydia/Gonorrhea mail-in kits (Simple 2)	Urine or vaginal swab (mail to lab)	Example: \$99 kit (LetsGetChecked). [40]	Home sample collection with lab NAAT; positives need treatment via clinician. [40]

Workflow note: For infectious disease and STI testing, the value of a home test depends on (1) correct sampling, (2) known test performance for the intended population, and (3) an unambiguous plan for confirmatory testing and treatment.

3.5 Saliva and breath testing

Saliva and breath tests are non-invasive but can be sensitive to timing, contamination, and device calibration. They are used in both medical and wellness contexts.

Test	Specimen	Measures	Typical cost	Notes
Saliva cortisol (lab)	Saliva (mail-in or clinic)	Cortisol level / diurnal patterns	\$50-\$200+ (varies)	Useful in select endocrine evaluations; many wellness uses are poorly validated.
Breath alcohol tester	Breath	Ethanol via fuel cell/semiconductor sensors	\$30-\$150+	Consumer devices vary; law enforcement devices use higher-grade calibration.
FeNO (fractional exhaled nitric oxide)	Breath	Airway inflammation marker	Usually clinic-based	Used in asthma management in some settings; not common as a consumer device.
Breath hydrogen tests	Breath	Carbohydrate malabsorption / SIBO proxies	Usually clinic-based	Protocol-heavy; interpretation depends on symptoms and prep.

4. Clinic-based labs and point-of-care testing

Clinical laboratories provide high analytical validity through controlled pre-analytics, calibrated instruments, and quality systems. Point-of-care testing (POCT) brings assays closer to the patient (clinic, urgent care, bedside) with faster turnaround, sometimes at the cost of narrower panels or higher per-test cost.

4.1 Venipuncture panels (blood draws)

Venous blood draws enable broad panels: electrolytes, kidney/liver function, lipids, inflammatory markers, hormones, and more. Many common tests are inexpensive cash-pay; specialized panels can be substantially more.

Test (example)	What it measures	Common technology	Typical cash-pay cost examples	Notes
CBC	Hemoglobin/hematocrit, WBC differential, platelets	Automated hematology analyzers	Often ~\$29 (example). [24]	Screens anemia, infection/inflammation signals; interpretation depends on context.
Basic/CMP metabolic panel	Electrolytes, glucose, kidney/liver markers (panel-dependent)	Photometry, ISE (ion-selective electrodes)	Often bundled; pricing varies widely	Consider fasting state and hydration.
Lipid panel	Total cholesterol, LDL, HDL, triglycerides	Enzymatic assays	Example ~\$59. [24]	Fasting not always required; depends on question.
HbA1c	Average glycemia over ~3 months	HPLC or immunoassay	Example ~\$39. [24]	Affected by some hemoglobin variants and altered RBC turnover.
TSH	Thyroid-stimulating hormone	Immunoassay	Example ~\$49. [24]	Follow up with free T4/T3 as appropriate.
PSA (screening)	Prostate-specific antigen	Immunoassay	Example ~\$69. [24]	Screening decisions depend on age/risk; false positives occur.
Vitamin D	25(OH)D level	Immunoassay or LC-MS/MS	\$72-\$99 (example range). [24]	Assay methods vary; avoid over-testing absent a clinical reason.

4.2 Cardiovascular clinic testing (common examples)

Cardiac assessment mixes biosignals (ECG), imaging (echo), and stress physiology.

Test	What it measures	Setting	Typical cost examples	Risks / limitations
Resting 12-lead ECG	Electrical activity across 12 leads	Clinic/ER	Often low cost; bundled in visits	Snapshot; may miss intermittent arrhythmias.

Test	What it measures	Setting	Typical cost examples	Risks / limitations
Holter monitoring (24-48h)	Continuous ECG	Clinic-prescribed	MDsave range ~\$120-\$378. [32]	Skin irritation; adherence matters.
Transthoracic echocardiogram	Structure/function; valves; flow (Doppler)	Imaging center	MDsave example ranges ~\$501-\$1,383 (regional). [31]	No ionizing radiation; operator and body habitus affect quality.
Exercise stress test	Ischemia signs and functional capacity under load	Cardiology lab	Varies widely	Requires exertion; rare adverse events.

4.3 Preventive screening tests (non-imaging examples)

Preventive screening is usually guideline-driven and age/risk dependent. Below are common specimen-based screenings and approximate out-of-pocket pricing examples.

Screening	What it measures	How it's collected	Typical cost notes	Notes
Cervical cancer screening (Pap ± HPV)	Cytology and/or HPV DNA	Pelvic exam sample from cervix	GoodRx notes that without insurance the cost is generally under \$150. [26]	Screening interval depends on age and test type.
PSA (prostate)	Prostate-specific antigen	Blood draw	GoodRx example ~\$69. [24]	Screening decisions depend on age/risk; false positives occur.
Hepatitis C antibody (and confirmatory RNA)	HCV exposure and active infection	Blood draw	Pricing varies; often recommended once for adults in many guidelines	Confirmatory RNA test needed if antibody positive.
HIV screening (lab)	HIV antigen/antibody	Blood draw	Pricing varies; high value in many contexts	Window periods apply; follow lab/CDC guidance.

5. Imaging technologies and screening imaging

Imaging provides structural and functional information that labs and wearables cannot. The major trade-offs are ionizing radiation (X-ray/CT/nuclear), contrast risks, incidental findings, and cost.

5.1 Modality overview

Modality	Physics	Best for	Radiation	Typical cost (cash examples)	Key safety notes
Ultrasound	High-frequency sound waves	Soft tissue, pregnancy, vascular flow (Doppler)	None	Varies; often hundreds	Operator-dependent; limited by gas/bone.
X-ray	Ionizing radiation projection imaging	Bone, lungs, certain foreign bodies	Low (e.g., adult chest X-ray ~0.1 mSv). [12]	Often <\$200-\$500 depending on setting	Minimize repeat studies; pregnancy considerations.
CT	Ionizing radiation + computed reconstruction	Trauma, lungs, abdomen/pelvis, vascular (with contrast)	Moderate to high (example: CT abdomen effective dose ~15 mSv). [14]	MDsave cites CT average ~\$500-\$1,000 (varies). [34]	Radiation exposure; iodinated contrast can cause reactions and kidney concerns in select patients. [16]
MRI	Magnetic fields + radiofrequency pulses	Brain/spine, joints, soft tissue, vessels (MRA)	None	MDsave MRI without contrast example ~\$440-\$951 (regional). [33]	Metal implants/foreign bodies can be dangerous; gadolinium contrast risks higher in severe kidney disease. [15,16]
Nuclear (PET/SPECT)	Radiotracers (gamma/positron)	Metabolic activity, oncology staging, perfusion	Yes (varies)	Often thousands	Radiation + tracer considerations; used when it changes management.

5.2 Radiation dose intuition

Radiation dose is commonly expressed as *effective dose* (mSv), a population-average measure used to compare procedures. RadiologyInfo notes natural background radiation in the U.S. is about 3 mSv per year. [13]

Exam (adult)	Typical effective dose (example)	Perspective
DEXA (bone densitometry)	~0.001 mSv (very low). [12,17]	Comparable to a day or less of background radiation
Chest X-ray	~0.1 mSv. [12]	Comparable to about 10 days of background radiation
CT abdomen	~15 mSv (example). [14]	Comparable to multiple years of background radiation

Dose varies by scanner, protocol, and patient size. Use imaging when it is expected to change decisions, and prefer the lowest-dose modality that answers the clinical question.

5.3 Point-of-care and handheld ultrasound

Handheld ultrasound turns a smartphone/tablet into an imaging console. It can be powerful in trained hands (cardiac, lung, abdominal, vascular checks), but it is operator-dependent and typically a clinician tool rather than a consumer diagnostic device.

Device category	Examples	Typical cost	What it enables	Notes
Handheld ultrasound (subscription model)	Butterfly iQ+	Probe pricing example ~\$2,299 plus membership plans (e.g., \$299/yr Core). [65,66]	Multi-system ultrasound exams with cloud workflow	Training and QA are crucial; image interpretation is the bottleneck.
Handheld ultrasound (probe-per-type)	Philips Lumify probes	Example listings show ~\$6,000 for a probe (varies). [67]	Higher-end handheld imaging with app-based console	Still operator-dependent; probe choice matters (linear/curvilinear/phased).

5.4 Screening and risk stratification imaging (selected examples)

Test	Modality	Typical cost examples	Radiation	Notes
Mammography (2D vs 3D tomosynthesis)	X-ray	GoodRx cites national averages: ~2D \$400+ and 3D ~\$658. [74]	Low (screening X-ray dose) [12]	3D may reduce callbacks in some settings; coverage varies.
Coronary artery calcium (CAC) score	Non-contrast CT	MDsave: prices start at \$35; varies by region. [71]	Low-to-moderate (protocol-dependent)	Used to refine ASCVD risk in selected patients; not for acute symptoms.
Low-dose CT lung screening	Low-dose CT	MDsave: \$99-\$418 (range). [72]	Lower-dose CT protocol	Used for lung cancer screening in eligible high-risk groups; guideline-driven.
Cardiac CT coronary angiography (CCTA)	CT + contrast	MDsave example regional range ~\$447-\$891. [73]	CT dose varies	Evaluates coronary anatomy; requires contrast and heart rate control in some protocols.

5.5 Elastography (non-invasive fibrosis assessment)

Elastography estimates tissue stiffness. In liver care, transient elastography (FibroScan) is used to estimate fibrosis/cirrhosis risk without a biopsy in many scenarios. [76]

Test	Technology	Typical cost examples	Invasiveness	Notes
FibroScan (liver)	Ultrasound-based transient elastography	MDsave: ~\$232-\$253 (cash-pay range). [70]	Non-invasive	Results depend on obesity, inflammation, and operator technique; interpret with labs and clinical context.

6. Functional diagnostics (cardiopulmonary, neuromuscular, pulmonary function, balance, cognition)

Functional tests quantify performance of integrated systems: how the body responds to stress (exercise), how nerves conduct signals, or how the lungs exchange gas. These tests often sit between wearables and imaging: more informative than consumer proxies, less invasive than many procedures.

6.1 Cardiopulmonary exercise testing (CPET) and VO2max

CPET measures gas exchange (oxygen uptake and CO₂ production) during graded exercise, alongside ECG, blood pressure, and ventilation. It can help explain unexplained dyspnea and assess cardiopulmonary limitations. [58]

Test	Technology	Primary outputs	Where performed	Typical cost notes
VO2max test (performance-oriented)	Mask + metabolic analyzer (portable or cart) during treadmill/cycle protocol	VO2max, ventilatory thresholds, exercise economy	Sports lab / wellness clinic	Common self-pay pricing examples range ~\$100-\$250 (provider-dependent). [60]
Clinical CPET	Metabolic cart + 12-lead ECG + BP + pulse oximetry; sometimes arterial line	Peak VO ₂ , ventilatory efficiency (VE/VCO ₂), limiting system	Hospital / specialty lab	Example: one provider lists \$500 self-pay for athletic performance CPET; clinical pricing varies. [59]

6.2 Neurology and neuromuscular testing

Neurophysiology tests measure electrical signals in the brain (EEG) and in peripheral nerves/muscles (NCS/EMG). They are typically outpatient procedures and are minimally to moderately invasive depending on needle use.

Test	What it measures	Invasiveness	Typical cost examples	Notes
EEG (awake/asleep)	Scalp electrical potentials (brain rhythms)	Non-invasive (surface electrodes)	MDsave range ~\$366-\$794. [57]	Used for seizure evaluation and other neurologic assessments.
Nerve Conduction Study (NCS)	Peripheral nerve conduction velocity/amplitude	Low (surface electrodes; brief shocks)	MDsave example: 9-10 nerves ~\$679-\$1,344. [56]	Often paired with EMG; interpretation requires context.
EMG (electromyography)	Muscle electrical activity	Moderate (needle electrodes)	MDsave range ~\$156-\$466 for one limb. [55]	Discomfort and minor bleeding/bruising possible.

6.3 Pulmonary function laboratory testing

Pulmonary function testing spans simple spirometry to full PFT labs that measure lung volumes and diffusion capacity. When oxygenation or acid-base status is central, an arterial blood gas (ABG) provides direct measurement but is more invasive.

Test	What it measures	Invasiveness	Typical cost examples	Notes
Spirometry (office)	FEV1, FVC, flow-volume loop	Non-invasive (breathing test)	MDsave: ~\$192-\$258 (in-office example; varies). [79]	Technique-dependent; requires maximal effort.
Pulmonary Function Test (PFT) panel	Spirometry + lung volumes ± DLCO	Non-invasive	MDsave: ~\$320-\$924 (cash-pay range). [77]	More complete characterization of restrictive vs obstructive patterns.
Arterial blood gas (ABG)	PaO2, PaCO2, pH, HCO3-, O2 content	Moderate (arterial puncture)	MDsave: prices start ~\$197 (varies). [78]	Gold standard for gas exchange/acid-base; can be painful and operator-dependent.

6.4 Balance, mobility, and cognition

These domains are often best measured with a combination of performance tasks and questionnaires. In research and high-end clinics, motion capture and force plates provide detailed biomechanics; in home settings, smartphones and wearables provide coarse but frequent metrics.

Domain	Representative measures	Tools	Notes
Mobility	Gait speed, step variability, sit-to-stand	Wearables + standardized timed tests	Gait speed is a strong global function marker; standardize footwear and environment.
Balance / vestibular	Postural sway, head impulse metrics	Force plates, inertial sensors, clinic vestibular tools	Useful for fall-risk and dizziness workups; usually clinic-based when symptomatic.
Cognition / mood	Memory, attention, executive function; depression/anxiety scales	Neuropsych testing batteries; validated questionnaires	Consumer cognitive apps can track trends, but formal testing is needed for diagnosis.

7. Invasive diagnostics (endoscopy, biopsy, lumbar puncture, catheterization, implants)

Invasive diagnostics provide high-resolution information (direct visualization, tissue histology, compartment-specific fluids) but introduce procedural risks and higher costs. They are typically justified when results can materially change management or when less-invasive tests are inadequate.

7.1 Endoscopy and colonoscopy

Endoscopic procedures use a flexible scope with a camera to directly visualize mucosa and, when needed, take biopsies or remove polyps. Colonoscopy can be both diagnostic and therapeutic (polyp removal).

Procedure	What it measures/does	Typical cost examples	Key risks	Notes
Colonoscopy	Visualizes colon; can biopsy/remove polyps	MDsave: \$1,243-\$4,142 (cash-pay range). [28]	Bleeding, perforation; sedation risks	Mayo Clinic lists bleeding/perforation among risks; perforation is rare but serious. [18,19]
Upper endoscopy (EGD)	Visualizes esophagus/stomach/duodenum; biopsy possible	MDsave: \$990-\$3,983 (cash-pay range). [61]	Bleeding/perforation (rare); sedation risks	Used for evaluation of GI symptoms, ulcers, Barrett's, etc.

7.2 Biopsy and tissue diagnosis

Biopsies obtain tissue for histopathology (and sometimes genomics/proteomics). They can be superficial (skin) or image-guided (breast, liver, lung).

Procedure	Sample	Typical cost examples	Risks	Notes
Skin biopsy	Skin lesion tissue	Often office-based; pricing varies	Bleeding, infection (low)	High value for suspicious lesions.
Breast biopsy	Breast tissue core/aspiration	MDsave examples vary by region; procedure page notes prices start ~\$990. [62]	Bleeding, bruising, infection (low)	Often prompted by mammography/ultrasound findings.

7.3 Lumbar puncture and compartment sampling

Lumbar puncture (spinal tap) samples cerebrospinal fluid (CSF) for infection, inflammation, bleeding, and other neurologic disorders.

Procedure	What it measures	Typical cost examples	Risks	Notes
Lumbar puncture	CSF cell counts, protein/glucose, cultures, biomarkers	MDsave: \$838-\$2,073 (cash-pay range). [63]	Post-LP headache, bleeding, infection (rare)	Often paired with imaging when increased intracranial pressure is a concern.

7.4 Catheter-based cardiovascular diagnostics

Cardiac catheterization can directly measure pressures and visualize coronary arteries (angiography). It is typically reserved for higher-risk evaluations and interventional planning.

Procedure	What it measures/does	Typical cost examples	Risks	Notes
Coronary angiography (cardiac cath)	Direct coronary imaging with contrast; pressures	MDsave: \$3,755-\$7,758 (cash-pay range varies by region). [64]	Bleeding/hematoma, contrast reactions, rare stroke/MI	Higher fidelity than non-invasive tests; often leads to intervention if severe disease found.

7.5 Implantable and long-term monitors (selected examples)

Implantables extend monitoring duration and improve signal quality, but they require a procedure and carry infection/bleeding risks. Costs can be substantial and are highly system- and insurance-dependent.

Device / procedure	What it measures/does	Invasiveness	Cost notes	Risks / limitations
Implantable loop recorder (cardiac event recorder)	Long-term rhythm monitoring (months-years)	Minor surgical implant	MDsave example: ~\$11,388 (regional). [80]	Infection, bleeding, device migration; used for unexplained syncope/cryptogenic stroke workups.
Pacemaker generator insertion/replacement	Treats bradyarrhythmias; provides pacing + logs	Surgical procedure	MDsave example: ~\$30,515 (regional). [81]	Infection, lead complications; lifelong device management.

Device / procedure	What it measures/does	Invasiveness	Cost notes	Risks / limitations
Implantable CGM (Eversense-type)	Interstitial glucose via implanted sensor	Minor procedure for sensor insertion/removal	Pricing depends on insurance; manufacturer has offered \$99/6-month system for eligible individuals (program details apply). [82]	Requires procedure and calibration workflow; availability may vary.

8. Building a measurement stack (minimalist to advanced)

A practical way to cover 'whole health' measurement is to build a layered stack: start with low-burden, high-yield metrics collected frequently, and add targeted tests when risk, symptoms, or decisions justify escalation.

8.1 Minimalist stack (low cost, low burden)

- **Vitals at home:** validated upper-arm BP cuff (weekly or per protocol). [1-3]
- **Weight:** scale (daily/weekly) + waist circumference (monthly).
- **Activity and sleep:** wearable step count + sleep timing (trend-focused).
- **Basic labs (periodic):** CBC, CMP, lipid panel, HbA1c as risk-appropriate. [24]
- **Mental health:** brief validated screeners (e.g., PHQ-9, GAD-7) + sleep/stress diary.

8.2 Advanced stack (higher information density)

- **Continuous sensing:** CGM for time-in-range, variability, and intervention response. [20-23]
- **Arrhythmia capture:** smartwatch ECG or mobile ECG + patch monitor when intermittent symptoms occur. [5-9,32]
- **Body composition:** periodic DEXA for bone density / composition in select populations. [17,27]
- **Sleep physiology:** HSAT or lab polysomnography when apnea is suspected. [29,49]
- **Imaging:** targeted ultrasound/CT/MRI when the result is expected to change management; understand radiation and contrast trade-offs. [12-16,33,34]
- **Targeted screening imaging:** mammography, CAC scoring, or low-dose CT when guideline-eligible. [71,72,74]
- **Operator-dependent tools:** handheld ultrasound in trained hands. [65-67]

8.3 Escalation triggers and safety

A measurement is only as good as the response plan. Common escalation patterns:

- Abnormal or inconsistent home readings -> repeat with standardized technique -> confirm with validated clinical testing.
- Concerning symptoms (chest pain, neurologic deficits, severe dyspnea) -> emergency evaluation rather than additional home testing.
- Positive screening tests (FIT, STI screens, wearable arrhythmia flags) -> confirmatory diagnostics and clinician follow-up.
- Imaging and invasive testing should be risk-stratified; avoid 'shotgun' testing that increases incidental findings without improving outcomes.

When stakes are high, prioritize devices/tests with transparent validation, appropriate regulatory status, and clear clinical pathways.

9. Emerging and controversial measurement technologies

Some measurement technologies are commercially available but have uncertain clinical utility in general populations. They can be appropriate in research, high-risk cohorts, or shared decision-making scenarios - but they warrant skepticism.

9.1 Multi-cancer early detection blood tests (MCED)

MCED tests aim to detect multiple cancers from a single blood draw using cell-free DNA or other biomarkers. Galleri (GRAIL) is one widely marketed example, available by prescription as a lab-developed test; the list price is \$949. [42]

Evidence is evolving. Recent reporting notes that large trials have not yet shown definitive population-level benefits such as mortality reduction, and sensitivity for early-stage cancers can be limited. [43]

Example	What it measures	Cost	Potential benefits	Key concerns
Galleri (blood test)	Cancer-associated signals + predicted tissue of origin	\$949 list price (discounts may exist). [42]	May detect some cancers lacking routine screening tools	False positives/negatives; follow-up imaging/biopsies; unclear mortality benefit for broad screening. [43]

9.2 Microbiome tests and personalized nutrition algorithms

Consumer microbiome sequencing and algorithmic nutrition plans are widely sold. They can be useful for education and hypothesis generation, but the mapping from microbiome composition to actionable interventions with proven health outcomes remains an active research area.

When evaluating these products, look for: (1) transparent methods (16S vs shotgun, QC), (2) reproducibility, (3) evidence that recommendations improve outcomes in controlled studies, and (4) a plan for follow-up or retesting.

References

References support key claims (device capabilities, costs, and safety considerations). URLs accessed in early March 2026; availability and pricing may change.

- [1] American Heart Association. Home Blood Pressure Monitoring. Updated Aug 14, 2025. <https://www.heart.org/en/health-topics/high-blood-pressure/understanding-blood-pressure-readings/monitoring-your-blood-pressure-at-home>
- [2] American Heart Association. Home Blood Pressure Measurement Instructions (PDF). https://www.heart.org/-/media/files/health-topics/high-blood-pressure/measuring_blood_pressure_updated_081024.pdf
- [3] ValidateBP.org. Validated Blood Pressure Devices (device list and guidance). <https://www.validatebp.org/>
- [4] U.S. Preventive Services Task Force. Hypertension in Adults: Screening. Apr 27, 2021. <https://www.uspreventiveservicestaskforce.org/uspstf/recommendation/hypertension-in-adults-screening>
- [5] Apple Support. Take an ECG with the ECG app on Apple Watch. <https://support.apple.com/en-us/120278>
- [6] U.S. Food and Drug Administration (FDA). De Novo classification letter for ECG App (DEN180044). Sep 11, 2018. https://www.accessdata.fda.gov/cdrh_docs/pdf18/den180044.pdf
- [7] Apple Newsroom. ECG app and irregular heart rhythm notification available today on Apple Watch. Dec 6, 2018. <https://www.apple.com/newsroom/2018/12/ecg-app-and-irregular-heart-rhythm-notification-available-today-on-apple-watch/>
- [8] AliveCor. KardiaMobile EKG Monitor. <https://kardia.com/products/kardiamobile>
- [9] AliveCor. Personal EKG devices pricing (KardiaMobile, KardiaMobile 6L). <https://store.kardia.com/products/personal-ekg-devices>
- [10] U.S. Food and Drug Administration (FDA). Pulse Oximeters (information and limitations). Jan 6, 2025. <https://www.fda.gov/medical-devices/products-and-medical-procedures/pulse-oximeters>
- [11] U.S. Food and Drug Administration (FDA). FDA announces additional recommendations to improve pulse oximeter performance across skin tones. Jan 6, 2025. <https://www.fda.gov/news-events/press-announcements/fda-announces-additional-recommendations-improve-pulse-oximeter-performance-across-skin-tones>
- [12] RadiologyInfo.org (ACR/RSNA). Radiation Dose from X-Ray and CT Exams. <https://www.radiologyinfo.org/en/info/safety-xray>
- [13] RadiologyInfo.org (ACR/RSNA). How much radiation dose do I get from different imaging procedures? https://www.radiologyinfo.org/en/info/safety-hiw_07
- [14] RadiologyInfo.org (ACR/RSNA). What is radiation dose? <https://www.radiologyinfo.org/en/info/safety-xray>
- [15] RadiologyInfo.org (ACR/RSNA). MRI Safety. <https://www.radiologyinfo.org/en/info/safety-mr>
- [16] RadiologyInfo.org (ACR/RSNA). Patient Safety: Contrast Material. <https://www.radiologyinfo.org/en/info/safety-contrast>
- [17] International Atomic Energy Agency (IAEA). Radiation protection of patients during DXA (bone mineral densitometry). <https://www.iaea.org/resources/rpop/health-professionals/other-specialities-and-imaging-modalities/dxa-bone-mineral-densitometry/patients>
- [18] Mayo Clinic. Colonoscopy. Updated Feb 6, 2026. <https://www.mayoclinic.org/tests-procedures/colonoscopy/about/pac-20393569>
- [19] Lohsiriwat V. Colonoscopic perforation: incidence, risk factors, management and outcome. World J Gastroenterol. 2010. <https://pmc.ncbi.nlm.nih.gov/articles/PMC2811793/>
- [20] GoodRx. Dexcom G7 15 Day prices, coupons & savings tips. Accessed 2026. <https://www.goodrx.com/dexcom-g7-15-day>
- [21] GoodRx. How much does Dexcom G7 cost without insurance? Dec 22, 2025. <https://www.goodrx.com/health-topic/diabetes/how-much-does-dexcom-g7-cost>
- [22] GoodRx. FreeStyle Libre 3 prices, coupons & savings tips. Accessed 2026. <https://www.goodrx.com/freestyle-libre-3>
- [23] Reuters. Abbott issues device correction for certain FreeStyle Libre 3 sensors over incorrect low blood sugar readings. Nov 24, 2025.

- [24] GoodRx. How much does blood work cost without insurance? Jul 22, 2024.
<https://www.goodrx.com/health-topic/procedures/how-much-does-blood-work-cost>
- [25] GoodRx. How much does a mammogram cost? Accessed 2026.
<https://www.goodrx.com/conditions/breast-cancer/how-much-mammogram-cost>
- [26] GoodRx. Pap smears: how often should I be getting them? (includes typical cost without insurance). Apr 1, 2022.
<https://www.goodrx.com/conditions/cervical-cancer/pap-smear-how-often>
- [27] MDsave. DEXA Scan (Bone Density Scan) - costs range from \$64 to \$432.
<https://www.mdsave.com/procedures/dexa-scan-bone-density-scan/d48afd>
- [28] MDsave. Colonoscopy - costs range from \$1,243 to \$4,142.
<https://www.mdsave.com/procedures/colonoscopy/d783fdcd>
- [29] MDsave. Sleep Test Home Study - costs range from \$146 to \$620.
<https://www.mdsave.com/procedures/sleep-test-home-study/d784ffc4>
- [30] MDsave. Sleep Study (Polysomnography) - example regional pricing.
<https://www.mdsave.com/procedures/sleep-study-polysomnography/d782f4c8/oklahoma/oklahoma-city-ok>
- [31] MDsave. Echocardiography with or without Color Doppler - typical pricing ranges (varies by region).
<https://www.mdsave.com/procedures/echocardiography-with-or-without-color-doppler/d784fbc8>
- [32] MDsave. Holter Monitoring - costs range from \$120 to \$378.
<https://www.mdsave.com/procedures/holter-monitoring/d783f8cc>
- [33] MDsave. MRI without Contrast - example regional pricing (\$440 to \$951 in Oklahoma).
<https://www.mdsave.com/procedures/mri-without-contrast/d781f5c9/oklahoma>
- [34] MDsave. CT scan cost overview (cites Healthcare Bluebook averages: CT ~\$500-\$1,000; MRI ~\$1,200-\$3,000).
<https://www.mdsave.com/t/imaging-radiology/ct-scan>
- [35] FDA. At-Home OTC COVID-19 Diagnostic Tests. Updated Feb 23, 2026.
<https://www.fda.gov/medical-devices/coronavirus-covid-19-and-medical-devices/home-otc-covid-19-diagnostic-tests>
- [36] iHealth Labs. iHealth COVID-19 Antigen Rapid Test (pricing example). Accessed 2026. <https://ihealthlabs.com/>
- [37] Walmart. OraQuick At-Home Rapid HIV Self-Test listing (pricing varies by retailer). Accessed 2026.
https://www.walmart.com/browse/health-medicine/oraquick-hiv-test/976760_1005860_542089_6724914_6007135
- [38] Reuters. U.S. FDA approves first at-home syphilis antibody test. Aug 16, 2024.
- [39] NOWDiagnostics. Press release: First To Know Syphilis Test available to consumers (pricing under \$30). <https://nowdx.com/blogs/news/first-and-only-in-home-test-for-syphilis-now-available-to-consumers-through-major-retailers-and-public-health-nowdiagnostics-launches-first-to-know%C2%AE-syphilis-test-across-the-u-s>
- [40] LetsGetChecked. Home STD testing pricing (Simple 2 and other panels).
<https://www.letsgetchecked.com/home-std-test/>
- [41] GoodRx. Does insurance cover at-home colon cancer screening tests? (includes FIT mail-in price example). Mar 23, 2022. <https://www.goodrx.com/conditions/colon-cancer/at-home-colon-cancer-test>
- [42] Galleri (GRAIL). Cost of the Galleri multi-cancer early detection test (list price \$949).
<https://www.galleri.com/patient/the-galleri-test/cost>
- [43] Wall Street Journal. Are multi-cancer blood tests ready for prime time? Published Mar 2, 2026 (reporting on trial outcomes and limitations).
- [44] U.S. Preventive Services Task Force. Colorectal Cancer: Screening. May 18, 2021.
<https://www.uspreventiveservicestaskforce.org/uspstf/recommendation/colorectal-cancer-screening>
- [45] Harvard Health Publishing. The future of blood pressure monitoring: cuffless devices. Oct 1, 2025.
<https://www.health.harvard.edu/heart-health/the-future-of-blood-pressure-monitoring-cuffless-devices>
- [46] Withings. BPM Vision (FDA-cleared home blood pressure monitor) pricing example.
<https://www.withings.com/us/en/bpm-vision/shop>
- [47] MobiHealthNews. Aktiia gets FDA clearance for OTC cuffless blood pressure monitor. Jul 7, 2025.
<https://www.mobihealthnews.com/news/aktiia-gets-fda-clearance-otc-cuffless-blood-pressure-monitor>

- [48] GoodRx. How do home sleep apnea tests work, and are they accurate? (includes cost ranges).
<https://www.goodrx.com/conditions/sleep-apnea/at-home-sleep-apnea-test>
- [49] GoodRx. How much does the average sleep study cost? Jul 12, 2022.
<https://www.goodrx.com/health-topic/procedures/how-much-sleep-study-cost>
- [50] GoodRx. Testing your A1C at home: what to know (cost starts around \$50 per kit).
<https://www.goodrx.com/conditions/diabetes/a1c-home-test>
- [51] GoodRx. Understanding the costs associated with diabetes (test strip cost estimates).
<https://www.goodrx.com/conditions/diabetes/supplies-for-diabetics>
- [52] GoodRx. Testing your A1C at home: what to know (home A1c kit cost).
<https://www.goodrx.com/conditions/diabetes/a1c-home-test>
- [53] GoodRx. At-home colon cancer screening tests (FIT example pricing).
<https://www.goodrx.com/conditions/colon-cancer/at-home-colon-cancer-test>
- [54] Cologuard. Cologuard Plus self-pay price information (\$790). <https://www.cologuard.com/what-is-cologuard-plus>
- [55] MDsave. Electromyogram (EMG) - one limb costs range from \$156 to \$466.
<https://www.mdsave.com/procedures/electromyogram-emg-one-limb/d786ffc4>
- [56] MDsave. Nerve Conduction Studies 9-10 nerves costs range from \$679 to \$1,344.
<https://www.mdsave.com/procedures/nerve-conduction-studies-9-10-nerves/d786ffcd>
- [57] MDsave. EEG (awake and asleep) costs range from \$366 to \$794.
<https://www.mdsave.com/procedures/eeg-awake-and-asleep/d482fbc4>
- [58] Cleveland Clinic. Cardiopulmonary Exercise Test (CPET) overview. Updated Nov 20, 2025.
<https://my.clevelandclinic.org/health/diagnostics/17221-metabolic-exercise-stress-test>
- [59] Crystal Run Healthcare. Cardiopulmonary Exercise Testing (CPET) - self-pay fee example (\$500).
<https://www.crystalrunhealthcare.com/services/cardiopulmonary-exercise-testing-cpet>
- [60] VO2 Master (industry blog). Portable metabolic testing FAQ (typical provider charges for VO2max). Apr 23, 2025.
<https://vo2master.com/blog/portable-metabolic-testing-faq/>
- [61] MDsave. Upper Endoscopy (EGD) costs range from \$990 to \$3,983.
<https://www.mdsave.com/procedures/upper-endoscopy-egd/d782ffc5>
- [62] MDsave. Breast Biopsy cost examples (varies by region). <https://www.mdsave.com/procedures/breast-biopsy/d78af5cf>
- [63] MDsave. Lumbar Puncture (spinal tap) costs range from \$838 to \$2,073.
<https://www.mdsave.com/procedures/lumbar-puncture-spinal-tap/d781ffc4>
- [64] MDsave. Cardiac catheterization with and without coronary angiogram costs range (varies by region).
<https://www.mdsave.com/procedures/cardiac-catheterization-with-and-without-coronary-angiogram/d785fec8>
- [65] Butterfly Network. Pricing and memberships for Butterfly handheld ultrasound (e.g., \$299/yr Core membership).
<https://www.butterflynetwork.com/pricing>
- [66] Butterfly Network. Special offers / probe pricing example for Butterfly iQ+ (e.g., \$2,299 with membership).
<https://www.butterflynetwork.com/offers>
- [67] Philips Lumify. Direct pricing for Lumify handheld ultrasound probes (example listings show \$6,000).
<https://philipslumify.com/collections/philips-direct-traffic>
- [68] Eko Health. DUO ECG + Digital Stethoscope FAQ (FDA cleared statement).
<https://support.ekohealth.com/hc/en-us/articles/13165630993691-DUO-ECG-Digital-Stethoscope-FAQ>
- [69] Eko Health. Stethoscopes and ECG-enabled digital stethoscope pricing (e.g., DUO/CORE).
<https://www.ekohealth.com/collections/stethoscopes>
- [70] MDsave. FibroScan (transient elastography) cost range (\$232-\$253).
<https://www.mdsave.com/procedures/fibroscan/d586f9ca>
- [71] MDsave. Cardiac CT Calcium Scoring cost (prices start at \$35; varies by region).
<https://www.mdsave.com/procedures/cardiac-ct-calcium-scoring/d785f4cf>

[72] MDsave. Low Dose CT Lung Screening cost range (\$99-\$418).

<https://www.mdsave.com/procedures/low-dose-ct-lung-screening/d785f5cf>

[73] MDsave. Cardiac CT Coronary Angiography cost example (regional range \$447-\$891).

<https://www.mdsave.com/procedures/cardiac-ct-coronary-angiography/d786ffc8/oklahoma/oklahoma-city-ok/73196?page=0>

[74] GoodRx. 3D mammogram breast screening: cost and comparisons (national average \$658; 2D average about \$400+).

Nov 5, 2024. <https://www.goodrx.com/conditions/breast-cancer/3d-mammogram>

[75] MDsave. CT scan cost overview (cites CT averages and includes related imaging categories).

<https://www.mdsave.com/t/imaging-radiology/ct-scan>

[76] WebMD. FibroScan: liver health assessment (explains transient elastography concept). Dec 16, 2024.

<https://www.webmd.com/fatty-liver-disease/fibroscan-assessment>

[77] MDsave. Pulmonary Function Test (PFT) cost range (\$320-\$924).

<https://www.mdsave.com/procedures/pulmonary-function-test-pft/d782facf>

[78] MDsave. Arterial Blood Gas (ABG) test cost (prices start at \$197).

<https://www.mdsave.com/procedures/arterial-blood-gas-test/d483f8cf>

[79] MDsave. Pulmonary Function Test (PFT) In Office cost range (\$192-\$258).

<https://www.mdsave.com/procedures/pulmonary-function-test-pft-in-office/d585ffcd>

[80] MDsave. Cardiac Event Recorder Implant cost example (Chicago): \$11,388.

<https://www.mdsave.com/procedures/cardiac-event-recorder-implant/d78bf5ca/illinois/chicago-il>

[81] MDsave. Insertion or Replacement of Pacemaker Pulse Generator (multiple leads) cost example (Bentonville, AR):

~\$30,515 (regional). <https://www.mdsave.com/procedures/insertion-or-replacement-of-pacemaker-pulse-generator-multiple-leads/d287fdc5/arkansas/fayetteville-ar/72712>

[82] Senseonics. Press release: savings offer for Eversense E3 CGM (eligible individuals pay \$99 per 6-month system). Mar

12, 2024. <https://www.senseonics.com/investor-relations/news-releases/2024/03-12-2024-110153489>